DO NOW — RECALL

Introduction to time and space complexity

Name: _____ Class: _____ Date: _____

1. What makes a binary search trace convincing?

2. State the average-case time complexity of quick sort for n items.

3. How many values are in an int array of length n?

4. Apply an eight-bit left shift by two to 00010110_2 and bitwise AND to 00110110_2 with 00001111_2. Give both results in binary.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Showing low, high, mid, comparisons and any bound updates.

2. Model answer

O(n log n).

3. Model answer

n values; storing it needs O(n) space.

4. Model answer

Left shift 00010110 << 2 = 01011000 (eight-bit, zero-filled). AND 00110110 ∧ 00001111 = 00000110.

Vocabulary — know these terms

time complexity — Asymptotic growth in algorithm work.

auxiliary space — Extra memory used by an algorithm.

Big O — Upper asymptotic bound commonly used to classify growth.

quadratic — Growth proportional to n^2.

exponential — Growth such as 2^n.

NEW LEARNING

Read, annotate and explain

Analyse the loops

One full loop runs n iterations; a pair of n-sized nested loops performs n^2 inner iterations.

Separate time and space

Big O time counts growth of work; auxiliary space measures extra storage beyond the original input.

Compare feasible scales

At n=16, exponential growth can be dramatically more expensive than linear growth. Big O omits constants and measured run time.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

For n=8, compare one loop visiting each item with two full nested loops.

Predict first

One scan has eight iterations, giving linear growth O(n).

Explain the answer

Nested loops perform 8×8=64 inner iterations, giving quadratic growth $O(n^2)$.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State the Big O of (a) constant assignment, (b) one n-element loop, (c) two full nested n-loops and (d) repeatedly halving an interval.

Question 2

A program copies an n-element array into a new n-element array. Give its time and auxiliary-space complexity and justify both.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A method uses an in-place $O(n^2)$ sort and another uses merge sort with $O(n \log n)$ time and $O(n)$ auxiliary space. Explain one situation favouring each.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

O(n) time necessarily means an algorithm uses O(n) additional memory.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.